

Market Liquidity and Institutional Trading

During the 2007-8 Financial Crisis

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August 2012

Abstract

This paper shows that institutional sell-side herding increased bid-ask spreads and liquidity risk during the 2007-8 financial crisis. Such an impact on liquidity is most pronounced in firms with large numbers of institutions that sold the same stocks, that is, have correlated trades. For the same reason, we find institutional investors with a dedicated, buy-and-hold, investment style to be the least likely to herd; their trading activity did not affect stock market liquidity during the crisis. Our results are robust to alternative explanations, different test specifications and consistent with recent theories highlighting the negative impact of institutional trading activity on market liquidity during a crisis.

JEL Classification: G01; G14; G20

Keywords: Institutional Herding; Institutional Count; Institutional Holdings; Market Liquidity; Financial Crises

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1. Introduction

Recent theories highlight the role of institutional trading in reducing market liquidity during a financial crisis (see Kyle and Xiong, 2001; Xiong, 2001; Garleanu and Pedersen, 2007; Brunnermeier and Pedersen, 2009; Huang and Wang, 2009). They predict institutions' sell-orders during negative extreme market shocks exacerbate order imbalances. In this paper, we corroborate these theories by presenting empirical results which suggest that institutional sell-side herding is a key factor in impairing market liquidity and creating liquidity risk. Our paper makes two important contributions: First, we show that the number of institutional shareholders is a (noisy but readily available) proxy for the level of institutional trading and the potential institutions' sell-side herding during a market downturn (Nofsinger and Sias, 1999; Sias, 2004; Zhou and Lai, 2009; Chiang and Zheng, 2010). Second, we illustrate the significant effect of institutional sell-side herding and correlated trades on both trading costs and liquidity risk during the 2007-8 financial crisis.

Shares of firms with a greater number of institutional shareholders are more likely to suffer from institutional herding when they exit at the same time, creating excess order imbalances and wider spreads. We do not equate herding to irrational behavior and do not make any attempt to discern which of the possible herding reasons drives our results. Recent theoretical studies suggest that increased selling activity during a crisis could be driven by several factors, such as increased funding constraints (Brunnermeier and Pedersen, 2009), higher risk aversion (Huang and Wang, 2009), and/or tighter risk management (Garleanu and Pedersen, 2007). All or some of the herding explanations could contribute to the herding results presented here. Companies with smaller numbers of institutional shareholders but large institutional holdings do not suffer as much, possibly because these institutions have better information about the firms, and are more likely to offload their shares in a more orderly manner to minimize the price impact of their transactions.

To further understand institutional characteristics and their trading patterns, we classify institutional investors into groups according to their 'type' and 'investment style'. Different investor types and styles have varying degrees of correlated trades both within and across their groups (Sias, 2004). Thus, given that a higher degree of correlated trades is associated with a greater impact on order imbalances, and hence illiquidity, we expect to find systematic differences in the impact of the various groups' trading activity on market liquidity. Based on the SEC 13f filings, institutional investors are classified into bank trusts

(BNK), insurance companies (INS), independent advisors (IA), public pension funds (PPS) and university and foundation endowments (UFE).¹ Furthermore, we use the database constructed by Bushee (2001) to group institutional investors into quasi-indexers (QIX), transient (TRA), and dedicated (DED) based on their long-horizon trading pattern.² In general, IA dominates our sample firms over the 2004-8 period, and over two-third of the institutional investor sector are quasi-indexers. We do not detect any significant correlation between ‘type’ and ‘style’; all three ‘styles’ appear in a similar sector representation within each ‘type’. UFE trades have high within-group standard deviation, are least correlated with the other institutional trading patterns, which diverge even further in the crisis period. The same applies to DED trades, which during the crisis became less correlated with QIX and TRA. This lower correlation in the trading pattern of UFE and DED with all the other groups leads to an insignificant impact of their herding activity on market liquidity. In contrast, the herding activity of all other institutional types and styles affects market liquidity. Separately, Koch et al. (2010) find a strong commonality among the Amihud price impact measure of stocks owned by mutual funds which have high turnover and liquidity shocks due to their investors’ withdrawals. Similarly, we find the impact of institutional trading on liquidity is the strongest among the Independent Advisor (IA) group, which includes mutual funds. Among the five institutional investor types, IA are the most likely to be subject to institutional constraints and the least able to stick to their trading strategies during a financial crisis, thus we expect them to have correlated trades. But unlike Koch et al. (2010), we find our results are statistically significant only for institutional sell-side trades during the crisis. The very high correlated buy-side herding before the crisis did not have any impact on liquidity or liquidity systematic risk.

¹ In SEC 13f, the five types of institutional investors are bank trusts (BNK), insurance companies (INS), investment companies (INV), independent investment advisors (IIA), and miscellaneous (MSC). We follow Bushee (2001) and merge INV and IIA into one group, i.e., investment advisors (IA). Bushee has also identified corporate pension funds (CPS), public pension funds (PPS), university and foundation endowments (UFE), which were classified in 13f as MSC. We use in our analysis PPS and UFE, but do not investigate separately (the scarcely represented) CPS and other MSC.

² Bushee (2001) groups institutional investors into three investment styles, viz. quasi-indexers (QIX), transient (TRA) and dedicated (DED), based on factor and cluster analyses on portfolio characteristics such as portfolio turnover, stability of holdings, block size, percentage ownerships in firms, average investment size. QIX are long-horizon but well-diversified investors. TRA are diversified but high portfolio turnover investors. DED are characterized as making large investments in their portfolio firms and having extremely low turnover. Bushee produced two versions of such classification; one assumes institutions can switch style and the other uses the dominant style for the entire estimation period. We use the version that omits short term switching since the analyses here focus on the time series difference in trading and holding patterns of particular institutional investor groups between pre-crisis and crisis periods. This version of classification produces less switching in investment style during the crisis and should work against us finding significant results.

We recognize that there might be spurious relations and omitted variables that might explain our results. Therefore, we run a battery of tests to try to alleviate such concerns. We include in our model specifications several market and accounting-based control variables, which the literature has identified as important determinants of trading cost and liquidity risk. In addition, we use industry and firm fixed effect specifications to account for omitted variables. In all our models, we cluster the standard errors at the firm level to control for time-series dependence, and add year dummies to capture the effect of cross-sectional dependence. We run regression models based on the changes in the variables instead of their levels. We also run separate tests to control for alternative explanations based on institutional blockholders, or the lack of retail investors. Our main findings remain the same.

The remainder of the paper is organized as follows. In Section 2, we position our paper alongside the previous literature and present our hypotheses. In Section 3, we describe our data sources, variable definitions and provide some descriptive statistics. In Section 4, we report our results. Section 5 provides some discussion and concluding remarks.

2. Previous literature and testable hypotheses

Institutional investors are more likely than individuals to herd, that is, follow each other's trades.³ This is because they can respond faster, they are subject to similar informational and regulatory environments, and their managers face heightened moral hazard problems, which makes them more reactive to bad news. Sias (2004) provides explanations for institutional herding.⁴ His analysis makes it clear that institutional herding is not (always) irrational. Sias' explanations though describe institutional trading behavior under normal market conditions, when trading is mainly information driven. Recent theoretical studies present alternative reasons why institutions herd and become liquidity traders during a

³ Two recent studies on retail investor trading behavior present some evidence of herding among retail investors as well (Kumar and Lee, 2006; Dorn et al., 2008). Still, the extensive literature on herding provides a list of explanations for institutional herding, whereas we know little about the sources of retail herding. Also, the relatively small aggregate equity holdings of retail investors over our sample period justify our focus on the impact of institutional trading behavior on market liquidity. Finally, Corwin and Lipson (2011) show that it is institutional herding and not retail herding that drive liquidity.

⁴ Informational cascades happen when Institutions infer information from each other's trades when information is noisy. Investigative herding occurs because Institutions follow the same signals. Reputational herding refers to the situation where Institutions choose to share the blame rather than taking independent decisions. Fads, or following the trend, refer to Institutions following the same investment strategies because they are considered novel. Characteristic herding refers to Institutions being attracted to securities with the same set of characteristics.

financial crisis. Brunnermeier and Pedersen (2009) suggest that adverse shocks to the value of collateral induce sell pressure by institutions who reach their risk limits and funding constraints, which in turn leads to market-wide liquidity shortages. Similarly, Garleanu and Pedersen (2007) argue that tighter risk management by institutions during market downturns reduces liquidity provision. Furthermore, Xiong (2001) shows that in the case of extreme return shocks, wealth effects cause convergence traders to liquidate their positions, abandon their usual trading strategies and trade instead in the same direction as noise traders, thereby amplifying the original shock. Kyle and Xiong (2001) show that this effect could cause prices of fundamentally unrelated assets to move together and could lead to contagion. Huang and Wang (2009) argue that liquidity demand is endogenous and liquidity shocks typically occur on the sell side and are of large amounts. They anchor their theory on the balance between the need to trade and the cost of trading. Costly trading means non-continuous trade and increased risk aversion when the need to trade arises as traders face the uncertainty of the trade not being executed. Risk aversion reduces the desire to hold risky assets, so the impact on order imbalance is asymmetric and is greater on the sell side. Moreover, at low risk levels, the cost of trading outweighs the need to trade, so all traders stay away from the market. Trading takes place only when the idiosyncratic shock and risk aversion are large and sell side strongly dominates buy side. This big order imbalance drives down market prices even in the absence of changes in the fundamentals or information asymmetry. It is clear from the above studies that one should expect sell-side correlated trades from institutional investors during a market crisis. We do not investigate in this paper, which of the above reasons leads to the selling orders. During a crisis, all of them might play a significant role in raising the selling needs of institutional investors.

To date much of the previous research has focused on the impact of institutional shareholdings, firms' fundamentals and information asymmetry to explain securities trading costs. Given the above recent studies and the large presence of institutional investors in the financial markets, one may expect institutional trading activity itself to have an impact on trading costs. Demsetz (1968) and Benston and Hagerman (1974) long argue that the number of shareholders proxy for the number of share transactions, since it is positively related to the number of potential buyers and sellers of the stock. Here, we use the number of institutional shareholders (*II_count*) to proxy not only for the level of institutional trading activity but also the impact of this activity to market liquidity. There is no a priori predicted relationship between *II_count* and trading cost. However, the larger the *II_count*, the greater the potential

for the herding effect which will lead to order imbalances, and higher bid-ask spreads. In addition, we follow Lakonishok et al. (1992) and calculate a measure of institutional herding, which allows us to directly test this conjecture.

Consider the following transactions:

<u>Case</u>	<u>Number of institutions selling, #S</u>	<u>Number of institutions buying, #B</u>	<u>'Herding' #B/(#B + #S)</u>
1	9	1	0.1
2	1	1	0.5
3	1	9	0.9
4	90	10	0.1

For simplicity, we assume institutions sell to other institutions instead of individuals. This is not an unrealistic assumption given that institutions accounted for 75% of shareholdings before the crisis, a figure which rose to 78% during the crisis. With this assumption, all four cases will lead to no change in aggregate institutional holdings, which is defined as the proportion of a company shares held by institutional investors. Indeed, we cannot make any statement about whether or not there is an increase or decrease in the number of institutional shareholders after these transactions. If all the seller-institutions did not sell all their holdings and the buyer-institutions are new shareholders there will be an increase in the number of institutional shareholders, even for Cases 1 and 4. ' $\#B/(\#B + \#S)$ ' is what we will define formally later as a measure of herding. A number below 0.5 indicates there are more sellers than buyers. The herding measure however cannot differentiate between Case 1 and Case 4. Case 4 can take place only if the firm has a large number of institutional shareholders before the sell orders were made. Thus, it is important to use *II_count* in our analyses. Following the theoretical studies mentioned above, we expect sell-side herding (Cases 1 and 4) to have an impact on market liquidity during a crisis. We expect Case 4 to produce a greater impact on the liquidity of the stock, compared to Case 1. The greater the number of sellers, the less likely it is for them to be able to coordinate an orderly disposal. So following Huang and Wang (2009) and Brennan et al. (2012), we argue that when there is an order imbalance, it is the sell orders (i.e., herding measure smaller than 0.5) by a large number of institutions (as in Case 4) that create illiquidity and produce a larger impact on spread. We are agnostic about the impact of weak (Case 2) or buy-side (Case 3) herding on market liquidity during a crisis. Finally, we further argue that Case 4 produces a

greater impact on liquidity when the sellers belong to ‘type’ and ‘style’ groups that have higher (across group) correlations of trades.

During a financial crisis, liquidity traders (with sell-orders) place largely non-discretionary orders, even though they know that their selling activity will create a price impact. They may have varying degrees of urgency to sell stocks within some set time-limit. Some, but not all, of these liquidity traders may be able to choose which stocks to sell. If so, they first sell the most liquid stocks in their portfolios (Scholes, 2000; Anand et al., 2010). The resulting sell-orders and correlated trades lead to excess order imbalances in previously liquid stocks. Thus, this trading activity leads to higher increases in trading costs for (previously) liquid stocks bringing them closer to the average stock in the market and increasing the average trading cost during the crisis. So, during a financial crisis, all stocks become less liquid and liquidity risk (i.e., liquidity R-squared and liquidity beta) goes up.

Different institutions may trade differently depending on their liquidity constraint and trading style. When there is no liquidity constraint, a dedicated (DED) investor is not likely to herd and sell during the financial crisis. We have several anecdotal examples on how different types of institutions traded during the financial crisis. The University Foundation Endowment (UFE) sector in particular suffered severely due to their rigid spending plans and heavy investment in illiquid assets that suffered huge price declines during the crisis. UFE are typically having longer investment horizons. For this reason, they can provide liquidity to the market, and then reap higher returns as a premium in the long run. It is therefore plausible that during the crisis they had to sell shares that had the larger bid-ask spreads leading to particularly strong losses in endowment value. This means the trading pattern of DED and UFE will be less correlated with the institutional sector as a whole and may have exerted less impact on liquidity and liquidity risk. On the other hand, Koch et al. (2010) report that mutual funds face the same funding shocks and customers redemption demands. They find that the liquidity of shares held by mutual funds tends to move together and in the same direction. There are other explanations for co-movement of liquidity; Sias (2004) suggests that bank trusts resort to herding to minimize litigation risk, that is, they follow the trades of other investors so as to make it easier to defend their investment choices. We expect these considerations to increase not only within but across group trading correlation as well, for example between bank trusts and pension funds since both groups face heightened fiduciary responsibilities. Overall, different institutional investor types and styles are expected to have varying degrees of herding propensity. We explore this by also investigating separately the

impact of the trading behavior of each investor classification. We expect the impact of this heterogeneity in trading behavior on market liquidity to be significant. Given that the equity of the average firm is in the hands of different institutional investor groups, the more correlated the across group herding activity, the greater its impact on order imbalances, hence market illiquidity. This will also lead to higher co-movements in spreads and a greater liquidity risk.

3. Data

3.1 Sample Firms

We identify all the constituents of the S&P 1500 index, that is, the union of S&P Mid-Cap 400, S&P 500 and S&P Small-Cap 600, for the years 2004 to 2008. This index covers about 90% of the US market capitalization. We choose 2004 as the beginning of our sample period since it is the first full year after the implementation of the Sarbanes-Oxley Act (SOX) in 2003. The recent literature suggests that the enactment of SOX in 2002 has affected firms in various ways, for example by increasing disclosure levels but at the same time increasing the costs of compliance and reporting (e.g., Ahmed et al., 2010). Therefore, we expect institutional investment preferences to change after the enactment of SOX, and so concentrate on the post-SOX era to avoid biasing our results with the effect of an exogenous event. The choice of 2004 also helps us to create comparable pre-crisis and crisis periods (2004-6 vs. 2007-8).⁵

We use CUSIP as a unique firm identifier that allows us to merge the information from different databases.⁶ We delete from our sample firms that have multiple CUSIPs or have changed their CUSIPs during the sample period. Also, so as to avoid data matching problems across the different databases, our final sample includes only December fiscal year

⁵ We avoid providing a definition of what is a crisis and when did the crisis start. We simply follow Fahlenbrach and Stultz (2011) and classify 2007-8 as the crisis period. One could argue that the early part of 2007 is not a crisis period. The fact that we conduct our analyses at an annual frequency does not allow us to split 2007 in different sub-periods. But the classification of the earlier part of 2007 as a crisis period should work against us finding significant relationships.

⁶ The CUSIP (Committee on Uniform Security Identification Procedures) identifier is a unique code for companies and issues and is used extensively across different financial databases. CUSIP has three types: 6-digit, 8-digit and 9-digit. CUSIP 6 is a company identifier. CUSIP 8 is a stock/security identifier. CUSIP 9 has an extra check digit, which allows checking for consistency across the CUSIP formats. Normally, the first 6 and 8 digits of CUSIP 9 are CUSIP 6 and CUSIP 8 respectively, except for those companies dropped from our database, as explained in the main text.

end firms, which account for 65% of the S&P 1500 firms. We cannot foresee any reasons why this restriction would bias our results in favor of our predictions. To identify the firms' fiscal year ends, we use Compustat's "datadate" variable. Missing values in this variable and the exclusion of non-December year end firms leaves us with 1,276 unique firms and an unbalanced panel of 5,591 firm-years.

We access the following databases to retrieve our data: CRSP for daily firm market information, including stock prices, closing bid and ask prices, number of shares outstanding, share volume and stock returns; Compustat North America for annual accounting information, such as book value of assets and debt, as well as number of shareholders, as reported by the firm; Thompson Reuters 13f for institutional ownership characteristics; I/B/E/S consensus files, accessed through Datastream, provide information on analyst coverage.

3.2 Variable Definitions

In this Section we describe the key institutional variables used in this study, our proxies for stock market liquidity and commonality in liquidity, as well as control variables identified in previous studies as important determinants of liquidity and commonality.

Institutional Investors and stock liquidity

The extensive literature on institutional investment highlights the importance of the effect of institutional ownership on a firm's market liquidity (e.g., Sarin et al., 2000; Dennis and Weston, 2001; Agarwal, 2007; Cao and Petrasek, 2012). We follow the extant literature and define institutional ownership (*Institutional holdings*) as the ratio of the number of shares held by institutional investors to the total number of shares outstanding. The Thomson Reuters 13f database reports institutional holdings on a quarterly basis. Since all our sample firms have December fiscal year-end, we use the institutional ownership information for the last quarter. Demsetz (1968) and Benston and Hagerman (1974) long argued that the breadth of shareholder base proxy for the share trading activity. Thus, following Agarwal (2007), we

use as a proxy for institutional trading the number of institutional investors holding a firm's equity (*II_count*).⁷

Different types of institutions such as bank trusts, insurance firms, independent advisors, pension funds and endowment funds, have distinct investment preferences because of differences in their fiduciary responsibilities (Chung et al., 2010). The extant literature also classifies institutional investors according to their investment styles (see Bushee 1998, 2001). In order to further understand institutional characteristics and their trading patterns, we use the indicator variables developed by Brian Bushee.⁸ In terms of investment types, Bushee (2001) uses the 13f classifications and categorizes institutional investors into BNK, INS, IA, PPS and UFE (see footnote 1 for further details). Bushee also groups institutional investors into quasi-indexers (QIX), transient (TRA) and dedicated (DED) (see footnote 2 for further details). We merge this information with the 13f data to create separate holding and count measures for each institutional investor type and each investment style.

Institutional herding is defined here as the ratio of the number of buyers to overall number of institutional buyers and sellers. Using the SEC 13f data for each quarter, we count the number of institutions that increased (decreased) their holdings in a given firm and classify them as buyers (sellers) for calculating the herding ratio. Our herding measure is the average ratio across the four quarters in every fiscal year. Note that this herding measure is principally a 'head-count' measure and takes no consideration of the size of the trades. Since institutional investors need not liquidate all their holdings to be included in this head-count measure, it is possible for *II_count* and *Institutional holdings* to both increase whilst there is sell-side herding. Furthermore, Lakonishok et al. (1992), Sias (2004), and Chung and Zhang (2011) subtract from the herding measure the cross-sectional mean, that is, the expected proportion of buyers in the market. We do not follow this practice, given our empirical setting and in particular our interest in comparing the crisis with the pre-crisis period. Herding measures that are adjusted for the cross-sectional average could result in drawing the wrong conclusion, i.e., that there is no difference between the herding measure in the pre-

⁷ Indeed, *II_count* is positively and significantly correlated with share dollar volume, both in the crisis and the pre-crisis periods. In contrast, the correlation between institutional holdings and dollar volume is negative and insignificant for all periods. Furthermore, we record a significant correlation between *II_count* and II sell-side herding, which increased from 0.160 pre-crisis to 0.302 during the crisis. This is further indication that *II_count* captures II trading.

⁸ <http://acct3.wharton.upenn.edu/faculty/bushee/>, accessed in May 2010.

crisis and crisis periods. In Section 4, we also describe two variations to the herding measure that we calculate to test the robustness of our results.

As a proxy for a stock's market liquidity, we use the observed quoted spread. Goyenko et al. (2009) find that this daily liquidity measure is among the best when compared with its high frequency benchmarks.⁹ The annualized spread is calculated as

$$Spread_y = \frac{1}{D_y} \sum_{t=1}^{D_y} \frac{\max(Ask_t - Bid_t, 0)}{Mid_t}$$

$$Mid_t = \frac{Ask_t + Bid_t}{2} \quad (1)$$

where Ask_t and Bid_t are, respectively, the closing ask and bid prices on day t in year y , and D_y is the total number of trading days in year y . This is what is known as round trip spread. We did also calculate annualized sell-side and buy-side spreads but the correlations between the three measures are very high. Hence, we kept only the round trip spread.

In order to calculate liquidity commonality, we follow Chordia et al. (2000) and Brockman et al. (2009) and run the following time-series regressions using monthly observations for each firm in our sample:

$$\Delta Spread_{f,m} = \alpha + \beta_1 \Delta Spread_{M,m} + \beta_2 \Delta Spread_{M,m+1} + \beta_3 \Delta Spread_{M,m-1} + \delta_1 Return_{M,m} + \delta_2 Return_{M,m+1} + \delta_3 Return_{M,m-1} + \delta_4 \Delta Volatility_{f,m} + \varepsilon_{f,m} \quad (2)$$

where Δ denotes the proportional change in the variable between successive months. The information on the market (M) is calculated using the equal-weighted average (for Spread or Return) for all firms in our sample except firm f . Firm return volatility is calculated as the standard deviation using daily returns in month m . We run these time-series regressions separately for the years 2004-6 (pre-crisis) and 2007-8 (crisis). We define *Liquidity Beta* as the contemporaneous coefficient of the changes in the market portfolio spread (i.e., β_1). *Liquidity Beta* captures the systematic liquidity risk of a firm. We also use *Liquidity Adj. R²* as an alternative measure of liquidity commonality. This measure takes the values of the adjusted R²s from the firm-level time-series regressions in equation (2) and accounts for both the systematic and idiosyncratic variations (Brockman et al., 2009).

⁹ Goyenko et al. (2009) use effective spread, realized spread and price impact, calculated from intraday Trade and Quote (TAQ) and Rule 605 data as benchmarks when comparing low frequency estimates.

Control Variables

We use several control variables identified in the literature as determinants of stock market liquidity. It has been widely documented that the level of the stock price is related to a firm's market liquidity, since it controls for price discreteness and also acts as a proxy for firm risk, that is, low prices are associated with higher risk (Stoll, 2000). We estimate the annual stock price using the average of the daily closing prices. We also control for the volatility of daily returns, using an annualized figure of the standard deviation of the daily returns over a year. High volatility translates to high firm risk and high inventory costs for traders. At the same time, it is associated with a high level of information asymmetry between company insiders and outsiders, as well as between investors with different information sets (i.e., the adverse selection argument). For all these reasons, we expect to observe a negative relationship between return volatility and liquidity. High trading volume is associated with decreases in inventory risk (Stoll, 1978a,b; Hameed et al., 2010) and should therefore lead to higher liquidity, that is, lower spreads. Firm market performance, i.e., stock returns, can affect the firm's return volatility and therefore liquidity (Hameed et al., 2010), thus it is important to control for it. The daily return is averaged over the year and annualized.

In addition to these market-based control variables that are frequently used in the literature, we also examine the impact of several accounting-related firm characteristics. Firm size is expected to be positively correlated with market liquidity. Larger firms are more visible and more carefully scrutinized by both investors and analysts. An increase in information disclosure will render prices more efficient and this benefit is greater for large firms if they can attract institutional investors (Diamond and Verrecchia, 1991). Hence, the shares of larger firms are expected to be more liquid. We control for firm size by including in our models the value of total assets, retrieved from Compustat.¹⁰

Stock liquidity also interacts with capital raising behavior. Companies with liquid stocks prefer raising equity to debt (Lipson and Mortal, 2009). Lesmond et al. (2008) find that leverage-increasing activity increases bid-ask spreads and vice versa. Amihud and Mendelson (2008) explain that, the higher the company's leverage, the more sensitive is its value of equity to information asymmetry and adverse selection. This explains why leverage reduces stock liquidity leading to wider bid-ask spreads and higher price impact costs. We

¹⁰ We use the book value of assets, and not the market value of equity, as a proxy for firm size, to reduce the cross-correlation in our multivariate models. Using market capitalization does not alter our results.

measure financial leverage as the ratio of long-term debt to total assets, as reported in Compustat. Stock liquidity depends, among other things, on the shareholder base; the larger the base, the more liquid is the stock (see Amihud et al., 1999; Grullon et al., 2004; Lipson and Mortal, 2007). We capture the breadth of ownership by measuring the number of shareholders, as reported in the firms' annual reports in Compustat.

Another important parameter, which affects decision making within a firm and has a direct impact on firm performance, is shareholder rights. Managers who are protected from shareholders' actions against them are expected to be less efficient and less transparent. Bebchuk et al. (2009) construct an E-index (Entrenchment Index), which is based on six shareholder rights provisions. The higher the level of the E-index, the higher is the level of managerial entrenchment. Bebchuk et al. find that the E-index is better related to firm valuation than the G-index (which is an index based on 24 provisions, developed by Gompers et al., 2003). Managerial entrenchment is expected to have a negative relationship to market liquidity, since entrenchment leads to higher trading costs for shares if investors are deterred by the entrenchment provisions.¹¹ Finally, a key factor in the theory of liquidity through inventory control is the information asymmetry between the market maker and the investors, and the resulting adverse selection problems that face the market maker when he trades with an informed investor. One external influence that reduces information asymmetry is the role played by the security analysts. The larger is the number of analysts following the company, that is, the greater the analyst coverage, the more liquid is the company's stock (see Brennan and Subrahmanyam, 1995; Irvine, 2003). We measure analyst coverage using the number of next-year forecasts (FY1) issued by analysts. The number of year-one earnings forecasts (F1NE) is obtained from the I/B/E/S history file through Datastream.

3.3 Summary Statistics

Table 1 presents the summary statistics for our sample. Panels A, B and C provide, respectively, details about the distributions of the stocks' market liquidity, institutional investors and other firm characteristics for the pre-crisis period (2004-6), the crisis period (2007-8) and their differences. A Wilcoxon rank-sum test reveals that, for most of our variables, we can reject the hypothesis that the pre-crisis and crisis periods are drawn from the same distribution.

¹¹ We retrieve the information on the Entrenchment Index (E-Index) from Lucian Bebchuk's website: <http://www.law.harvard.edu/faculty/bebchuk/data.shtml>, accessed in May, 2010.

Consistent with expectations, Panel A shows that the average quoted spread increases from 0.157% to 0.235% during the 2007-8 financial crisis accompanied by an increase in the standard deviation. A similar pattern applies to the liquidity commonality, as measured by R^2 , with the mean value increasing from 6.9% to 41.5%. The Liquidity Beta has a different pattern, with a drastic decrease in cross-sectional standard deviation but a significantly higher average level in the crisis period.¹²

Panel B shows, on average, institutions hold 76% of our sampled firms' equity. The largest holders by far are independent advisors (IA) holding about half of all shares held by the institutional investor sector; IA are also large in number. The next in line are BNK, INS, PPS and UFE in that order (see footnote 1 for explanations of abbreviations). As they are also smaller in number, their average holdings are much larger than that of IA with the exception of UFE. UFE are universally small in number and small in proportional holdings. In terms of investment style, QIX have the largest holdings and head count, followed by TRA and DED in that order. Between the pre-crisis and crisis periods, institutional investors increased their holdings in the average S&P 1500 firm from 74.5% to 78.7%. Similar increases can be found for most institutional investor types and styles. Interestingly, the average number of institutional investors increases from 232 before the crisis, to 261 during the crisis. Thus, our subsequent findings on the effect of *II_count* on market liquidity are not driven by a reduction in the number of unique institutions we include in our analysis during the crisis.¹³

Panel C shows that the average firm in our sample is large, which is the result of our decision to study S&P 1500 firms. The average stock price is \$35 and the average book value of assets is just above \$20 billion. Over the entire sample period, the average stock return and return volatility are 2.9% and 39% respectively with a strong boom and bust pattern between the pre-crisis and crisis periods. Finally, the average firm has 20% leverage, just over 31,000 shareholders, approximately ten analyst forecast reports published about its one-year-ahead earnings and a 2.76 E-index, which reveals a tendency for more entrenched managers.

¹² The reduction in the variations of beta is a well known phenomenon as correlation with the market goes to 1. Nevertheless, scatter plots reveal sufficient variations in liquidity beta for it to produce a significant relationship with institutional variables, even when extreme values are removed.

¹³ Our sample covers 3,349 (3,097) unique institutions during crisis (pre-crisis). Thus, the reported results in this paper are not driven by a wave of defaults, M&As or divestment by institutions of their shares in the US equity markets.

Table 2, Panel A, compares the medians¹⁴ and standard deviations of our main herding measure for the pre-crisis and crisis periods. Conditional on a small standard deviation, a median herding measure that is above (below) 0.5 indicates a period of buy- (sell-) side herding. The sell-side herding of institutional investors increases during the crisis period across all types and styles.¹⁵ A K-sample test finds the differences to be statistically significant for all types and styles, except for DED. Interestingly, the cross sectional standard deviation of the herding measure dropped substantially during the crisis compared to the pre-crisis period for all groups with the exception of UFE. This indicates a substantial convergence of within group buy-sell pattern during the crisis. Panel B (C) presents the correlation coefficients across types/styles for the herding measure during the pre-crisis (crisis) period. The correlations are all higher in the pre-crisis period than in the crisis period. This indicates some asymmetry between buy-side and sell-side herding behavior, that is, the different groups of institutional investors buy together in good times but sell differently during crises. The IA group has the most highly correlated trading pattern, both within the group and with other groups. UFE trading pattern has the least in common within the group, as well as with the other groups.

Overall, our summary statistics bear three important facts for the S&P 1500 sample as a whole: average institutional holdings and count went up during the crisis; there was a switch from buy herding during the pre-crisis period to sell herding during the crisis; the between group herding correlation was stronger pre-crisis when buy side dominated than that during the crisis when sell side dominated. Since the institutional investor sector as a whole switched from buy side to sell side during the crisis, but at the same time there was an increase in *Institutional holdings* and *II_count*, the following explanations are all equally plausible: (i) institutional investors, as a sector, were net buyers during the crisis; (ii) institutional sellers reduced their positions but did not divest themselves completely of their shares; (iii) institutional buyers not only covered the positions of the sellers but also bought from some individual investors, leading to an increase in the net institutional holdings; (iv) there were new entrants or institutional investors became more diversified during the crisis

¹⁴ The mean values indicate a similar trend in most cases but in the pre-crisis period the means are driven by some clustering at the left tail of the distribution making some means smaller than medians and smaller than their counterparts in the crisis period.

¹⁵ In this paper, we use the quarterly 13f filings of S&P 1500 firms to determine changes in II ownership. This results in us not capturing all the net trades during the quarter, as well as short positions. These weaknesses in our data collection should bias our tests against finding significant results. Despite this, our herding measure clearly captures sell-side herding behavior among IIs during the crisis.

leading to an overall increase in *II_count* per firm. In the pre-crisis period, trading was strongly dominated by buy side and was highly correlated possibly because a lot of the buy-sell decisions were information driven. During the crisis, accurate information was scarce; the reduced trading correlation indicates that institutional investors' choices on which stocks to buy or sell diverged in this period. Nevertheless, the correlation of trading pattern at sell side turns out to be an important indication whether the *II_count* and institutional herding are to result in liquidity risk; for example, UFE and DED were the two groups with low correlated trades to the other institutional investor groups and exerted no impact on liquidity risk, as shown in the next Section.

Using quarterly data from 1983-1997, Sias (2004) documents that BNK exhibit the strongest evidence of herding. Our results for 2007-8 indicate that all institutions increased sell-side activities, with UFE experiencing the largest swing from buy to sell. INS, IA and PPS all have much larger swings than BNK. In Sias's sample, BNK are the largest institutional shareholders, compared to the other institutional investors, with an average head count of 189 (20% of total number of institutional investors). In our more recent sample period, the influence of BNK is much reduced and taken over by IA. During 2004-8, the number of average BNK per firm is 4 (which is less than 2% of total institutional investors per firm). BNK average holding is 14% which is about one-fifth of total institutional holdings. This might explain the different BNK herding behavior reported by Sias and this study.

4. Results

4.1 *Effect of II_count and Institutional holdings on market liquidity*

Table 3 presents our OLS regression results on the relation between market liquidity and institutional ownership. To alleviate concerns about endogeneity and spurious inferences, we include in all pooled-sample regressions the control variables¹⁶ that the literature identifies as important determinants of market liquidity. All model specifications include industry fixed effects to control for time-invariant omitted industry-level factors that affect

¹⁶ In order to reduce the impact of outliers we take the natural logarithm of the following variables that have skewed distributions: stock dollar volume, stock price, assets and number of shareholders. We also take the cumulative distribution function (cdf) of return volatility, instead of its level, to avoid multicollinearity issues. Indeed, a variance inflation factor (VIF) analysis for all of our model specifications reveals no significant problems, after these transformations.

liquidity.¹⁷ We also include year effects to control for cross-sectional dependence, that is, market-wide factors that affect a stock's market liquidity. Our reported p-values are based on heteroskedasticity-robust standard errors, clustered at the firm level. Clustering the errors at the firm level allows us to control for time-series dependence. We present separately the results for the crisis and pre-crisis periods as well as those with and without the accounting-based control variables.

In Table 3, the number of institutional investors is positive and significant in all specifications; it increases the quoted spread in both pre-crisis and crisis periods. This is consistent with our conjecture that the nature of institutional investors' trading behavior increases trading costs. Later on in our analysis, in Table 6, we provide direct evidence of this by illustrating that it is sell-side herding that has the main effect on trading costs. Consistent with Agarwal (2007), we also find that institutional holdings decrease quoted spreads.¹⁸

All the market-based control variables are highly significant and have the predicted signs, that is, stock returns and volatility are positively associated with spread, whereas stock dollar volume and stock price are negatively associated with it. In contrast, none of the accounting-based control variables appear to be significantly related to spread. Still, we keep them in the subsequent analyses since, in several specifications, some of these control variables appear to be marginally significant and increase the overall R^2 .

We then investigate whether the positive relationship between *II_count* and spread is sensitive to the institutional investor type and investment style. Table 3, Panel B, reports only the *II_count* and *Institutional holdings* coefficients, as well as the regression R^2 , from the

¹⁷ Falkenstein (1996) states that IIs seek liquidity. The significance of Institutional holdings on the right hand side of a regression explaining liquidity could therefore be due to an endogeneity bias. Agarwal (2007) performs a test of causality to investigate whether II ownership causes liquidity or if, instead, the relation is due to IIs seeking to invest in liquid stocks. He argues that the former holds.

¹⁸ The literature reports ambiguous results on the effect of Institutional holdings on spread. For instance, Dennis and Weston (2001) document a negative relation between the percentage of shares held by IIs and the log of the spread; their study is constructed in the same manner as ours. Sarin et al. (2000), on the other hand, find a non-significant relation between holdings and spread. Agarwal (2007) argues that the relation between spread and institutional holdings should be U-shaped. This relation arises from a tradeoff between IIs increasing the spread due to adverse selection and decreasing it due to the information efficiency brought about by informed trading. Differences in the estimation methods, sample periods, and data sources used in these studies may explain some of the differences between the various empirical findings, e.g., Agarwal reports average institutional holdings of 36%, which is less than half of that of our sample (i.e., 76%). Also, in our study the correlation between *II_count* and institutional holdings is low, 2.2%. It shows that the result on the effect of institutional trading on spread is not driven by multicollinearity. We also perform our analysis excluding institutional holdings and the positive, significant relation remains (untabulated result). Moreover, *II_count* cannot be a proxy for the squared institutional holdings; the pairwise correlation between these two variables is negative and only 4% in our sample. Still, we re-run our analyses including the squared institutional holdings and *II_count* to capture non-linearity and our results remain unchanged (untabulated result).

regressions we run for each institutional investor type and style. The regression model used is similar to that of Panel A, using the full set of control variables. The positive effect of *II_count* and the negative effect of *Institutional holdings* remain for almost all institutional investor types and styles. The only exception is for PPS during the pre-crisis period, but their effects become highly statistically significant during the crisis.

Next, we run a series of alternative model specifications to further examine the robustness of the results reported in Table 3. Table 4 reports two of them. Instead of examining the levels, we look at the annual changes of all variables. Regressions based on variable changes are typically less likely to capture spurious relationships. In columns 2 and 4, we also add the lagged change in the spread. This allows us to reduce autocorrelation in the residuals, caused by any non-stationarity in the time-series of spreads. Our results remain unchanged. Finally, we also run panel data regressions using firm fixed effects (untabulated results). The coefficient of *II_count* remains positive and highly significant (at the 1% level) in all specifications. Institutional holdings remain negatively related to spread, but they are highly significant (at the 1% level) only for the pre-crisis period (significance is at the 10% level during the crisis). We note that running firm fixed effects models when the time-series of the panel is short could introduce more bias into the coefficients compared to the OLS models. We therefore re-run our panel data regressions for the full sample period, using an indicator (crisis dummy) variable for the crisis period. The results remain unchanged. Finally, we interact all the independent variables with the crisis dummy to capture marginal effects. Consistent with the above results, the marginal and base effects of *II_count* on spread are positive and highly significant.

We now turn our attention to the relation between liquidity commonality and institutional ownership. To date, the literature on commonality in liquidity has identified firm size, industry effects and market conditions (see Chordia et al., 2000; Brockman et al., 2009; Karolyi et al., 2012; Rösch and Kaserer, 2012) as the main determinants of commonality. Watanabe and Watanabe (2008) also show that commonality in liquidity is related to periods of increased information asymmetry. Consistent with the prior analysis, we run OLS regressions and include industry and year fixed effects. We also control for firm size using the book value of the firm's total assets. We cluster the heteroskedasticity-robust standard

errors at the firm level. We run our analyses both for the systematic (beta) and overall (R^2) liquidity risk.¹⁹

Table 5 reports the results for the crisis period. We do not present the results for the pre-crisis period, essentially because the parameter estimates are largely insignificant. In panel A, we present the result for the overall *II_count* and *Institutional holdings*. Panel B reports only the coefficients for count and holdings from separate regressions per institutional investor type and style. In line with our prediction, *II_count* is positively and significantly associated with both the liquidity beta and R^2 . *II_count* for DED did not drive liquidity R^2 and has only weak impact on liquidity beta. We demonstrate in the next Section that *II_count* proxies for institutional investor sell-side herding, and provide evidence that DED was the least likely to herd in the crisis of 2007-8. Finally, there is no significant association between liquidity commonality and the level of institutional shareholdings (only for TRA is *Institutional holdings* negative significant in both models).

4.2 Impact of institutional herding on market liquidity

In this Section we illustrate that institutional herding is the reason behind the positive effect of institutional trading on trading costs and liquidity commonality during crises. Table 6 presents the results on the effect of institutional herding on quoted spread and liquidity risk. The model specifications used are identical to these in Tables 3 and 5, but instead of capturing *II_count* and *Institutional holdings*, we now use the institutional herding measure to capture the impact of the institutional investors' trading behavior on market liquidity.

The coefficient of institutional herding is negative and highly significant in most specifications. So, with the clear exceptions of UFE and DED, sell-side herding during the crisis increased quoted spread, liquidity R^2 and liquidity beta for most groups. It appears that the larger the sell-side herding by institutional investors of a firm's stock (the smaller the value of the institutional herding measure) the larger is the quoted spread, as well as liquidity R^2 and beta of the firm during the crisis. In the pre-crisis period, trading was dominated by buy side. All the liquidity R^2 and liquidity beta results for the pre-crisis period are statistically

¹⁹ Using scatter plots we identify liquidity risk and *II_count* values that could be considered as outliers. We run several sensitivity tests using different cut-off points for excluding observations. Our results are not driven by extreme values.

insignificant (untabulated result).²⁰ It appears that only strong sell-side herding affects liquidity risk. This could also explain the finding of Watanabe and Watanabe (2008) that high liquidity risk states of the economy are short-lived and characterized by heavy trading. During the crisis period, the herding measures for UFE and DED have the largest within group standard deviation and the smallest correlation with other groups. This explains the lack of results for these two groups. DED investors are, by definition, buy-and-hold investors, who engage with a firm for the long-term; they are not expected to have similar herding behavior to other institutions during a crisis. Indeed, the change in DED herding measure in Table 2 is not significant during the crisis. The herding behavior of institutional investors is much more pronounced for QIX and TRA investors.

Table 6 tests a linear relation between institutional investor herding and liquidity. But our predictions are based on sell-side herding only. In order to isolate the effect of sell-side herding we create two new variables: a sell-side herding dummy which takes the value of one when institutional herding is lower or equal to 0.5, zero otherwise, and a Degree of Imbalance (DOI) variable, which is calculated as the absolute value of the difference of institutional herding from 0.5. DOI captures the extent of herding imbalance in a firm. In Table 7, we re-run the analyses of Table 6 but now using the interaction of DOI with the sell-side herding dummy, as well as the uninteracted terms. In the full sample analysis, the coefficient of the interaction term is positive highly significant thus confirming that the greater the herding imbalance caused by sell-side herding the higher the spread and liquidity commonality. In line with previous findings, this result is mainly driven by IA and QIX. BNK, PPS and TRA also report positive coefficients of similar statistical significance. In contrast, the impact of sell-side herding on liquidity is unexpectedly negative for INS (in the first model where spread is the dependent variable; otherwise insignificant) and largely insignificant for UFE and DED, which is consistent with prior results.

Recently, Koch et al. (2010) document that correlated trading can also induce liquidity commonality. Our findings corroborate the findings of Koch et al. (2010) on the impact of mutual fund trading (in our classification included in the IA group) on liquidity commonality. However, our findings also highlight the significant impact of the trading activities of other institutional investors, such as BNK and PPS, during a crisis, as well as that

²⁰ The negative relation between institutional herding and spread exists even in the pre-crisis period, though the impact of sell-side herding on trading costs is smaller (untabulated). This is reasonable given the prevalence of buy-side herding in 2004-2006.

only correlated sell-side herding and not the correlated buy-side herding can have such an impact.²¹

4.3 Further robustness checks

We run five additional (untabulated) analyses, in order to further strengthen our inferences. First, an alternative explanation for our results could be related to the composition of ownership and, in particular, the lack of individual (retail) investors in firms with a high number of institutional investors. Amihud et al. (1999) argue that the higher the number of individual investors holding a firm's equity, the higher the noise trading, and hence the higher the market liquidity. In addition, they find that a reduction in the proportion of shares held by individual investors is associated with an increase in adverse selection costs, since the market makers have to deal with the remaining better informed investors. However, in our sampled firms, the number of institutional investors is positively, significantly, correlated with both the number of individual investors and their proportional representation. Our results do not appear to be driven by the lack of individual investors in high *II_count* firms. Our findings are consistent with the recent evidence of Corwin and Lipson (2011) that it is the correlated trades of professional and not retail traders that drive commonality in the prices and liquidity of NYSE listed stocks.

Second, if our conjectures regarding the role of *II_count* are correct, one would expect a greater impact of sell-side herding on liquidity in firms with a greater number of institutional investors. In order to test this, we create a dummy variable which takes the value one (zero) for any firms with above (below) median number of institutional investors during the crisis. We re-run the analyses of Table 6 but now also include an interaction term of the *II_count* dummy and institutional herding, as well as the uninteracted variables. This interaction term, which captures the marginal effect of institutional herding on liquidity in firms with high *II_count*, is negative and highly significant in all regressions. Thus, we confirm that the impact of sell-side herding on increasing illiquidity, i.e., higher spread and liquidity risk, is more pronounced in high *II_count* firms.

Third, firms with blockholders have lower free float and less liquid stocks (Rubin, 2007). Sell-side herding in these firms should result in greater order imbalances, hence we

²¹ This also corroborates the results of Brennan et al. (2012) who show that sell-side illiquidity is priced far more strongly than buy-side illiquidity.

expect greater impact of sell-side herding on liquidity in the presence of blockholders. We classify firms into two groups, that is, with or without a blockholder(s). We use, in separate analyses, indicator variables based on the holdings of the largest shareholder and the largest five shareholders. The median holdings of the largest blockholder was around 9% of the equity of the firm during the crisis. The median holdings of the five largest shareholders was 30% during the same period. Therefore, our indicator variables take the value one for values above 9% and 30% respectively, and zero otherwise. Following the analysis above, we interact the indicator variables with institutional herding to capture marginal effects. As expected, the marginal effect of institutional herding on spreads and spread commonality is larger in the presence of blockholders (highly economically and statistically significant).

Fourth, in order to test the robustness of our inferences regarding the importance of the trading behavior of different institutional investor groups, we follow Sias (2004) and run standardized regressions where all variables are transformed to have zero mean and unit variance.²² This standardization allows us to compare the sizes of the coefficients. The magnitude of the coefficients is in line with our inferences. For example, we observe that the herding by QIX investors has a more pronounced (similar) effect on liquidity beta (overall liquidity R^2) than (to) that of TRA investors. The coefficients of DED herding are very close to zero and several times smaller than those of QIX and TRA investors.

Fifth, we repeat the analysis of Table 6 using a dollar-weighted herding measure,²³ proposed by Bohl and Voronkova (2003). The coefficients for overall herding remain negative and significant but are weaker, further indicating that the head count measure has a stronger impact than the dollar volume weighted holdings in driving liquidity risk.

5. Discussion and Conclusion

This paper investigates the role of institutional investor trading in escalating market illiquidity during the recent financial crisis. We measure the intensity, scale and flow of institutional trading by the number of institutional investors holding shares, their proportional ownership and the ratio of institutions' buy trades to all trades. We study two aspects of market liquidity: trading cost as measured by the quoted proportional spread and

²² For each variable, both dependent and independent, we subtract the cross-sectional average and then divide by the cross-sectional standard deviation.

²³ Dollar herding is the ratio of the dollar value of shares bought to the dollar value of all shares transacted (bought and sold) over the period.

commonality in liquidity as measured by R^2 and the beta of the cross-sectional spread variations. Our results suggest that institutions' sell-side herding together with their correlated trades contributed to the increased trading costs and liquidity risk during 2007-8.

Our findings are robust to various different model specifications and measures. They provide empirical support for theoretical predictions on the role of trading in exacerbating liquidity shortages during a crisis. In particular, prior theoretical literature predicts the existence of increased selling activity by institutional investors during a crisis. The rationale is that, during a market downturn, institutional investors face high selling needs due to increased funding constraints (Brunnermeier and Pedersen, 2009), high risk aversion (Huang and Wang, 2009), tight risk management (Garleanu and Pedersen, 2007), or offloading of assets by convergence traders (Kyle and Xiong, 2001). A common prediction is that these pressures lead to excess order imbalances and inventory costs, hence greater illiquidity. In this paper, we provide evidence that firms with a larger number of institutional investors are more vulnerable to sell-side herding. We report that institutional investors did indeed increase their selling activity substantially during the financial crisis. More importantly, we show that this sell-side trading activity led to an increase in firms' trading costs and liquidity risk during the crisis. We do not investigate which of the above theoretical predictions drives our results. During a crisis, all of them might play a significant role in raising the selling needs of institutional investors.

Our results collectively indicate a significant role for institutional investors' sell-side herding in increasing a firm's trading costs and liquidity risk. They confirm the speculation by Chordia et al. (2000) that commonality in liquidity could be due to institutional herding (p. 14), and provide direct evidence to support the conjecture of Koch et al. (2010) that correlated trading increases commonality in liquidity. Our study extends Koch et al. by showing the impact of institutional herding on a firm's quoted spread, as well as demonstrating the prevalence of these phenomena across different institutional investor types during a crisis. Moreover, our results indicate that only few institutional investors, mainly these with a dedicated investment style, did not contribute to the increase in liquidity risk during the 2007-8 financial crisis. Controversial as it may sound, our findings also suggest that restricting the number of institutional shareholders holding a firm's equity might help to reduce the firm's liquidity risk during a financial crisis.

Scholes (2000) argues that, during a crisis, market participants first sell the most liquid stocks in their portfolios. Similarly, Anand et al. (2010) report that institutional

investors tilt their selling activity towards less liquidity-sensitive stocks during a crisis. These findings imply that our results for *II_count* and institutional investors' sell-side herding cannot be driven by reverse causality. Still, we accept that the endogeneity issues cannot be completely resolved. For example, one may argue that institutional investors sold stocks in anticipation of a liquidity drain out. Until we can successfully find an instrument to control for such explanation, the endogeneity problem cannot be eliminated.²⁴ But, our results indicate a strong connection between sell-side herding, *II_count* and liquidity risk even if the direction of causality remains as a conjecture. It is the first empirical evidence that is in line with the theoretical predictions of Xiong (2001), Brunnermeier and Pedersen (2009), Garleanu and Pedersen (2007) and Huang and Wang (2009).

In 2007, institutional investors held 80% of the S&P 1500 equity and accounted for the majority of the trading volume in the markets. Given the size of institutional investors' participation in the economy, institutional investors' trading behavior has a large impact on liquidity and trading costs and can easily induce market-wide systemic liquidity risk. So this paper not only provides important policy implications but also some empirical support to theoretical models of extreme liquidity (e.g. Huang and Wang, 2009; Kyle and Xiong, 2001), in particular regarding the role of (institutional investors') trading activity in exacerbating the liquidity drain-out during the recent financial crisis.

²⁴ Following recent studies, e.g., Faccio et al. (2011) and Laeven and Levine (2009), we calculate the cross-sectional average *II_count* of the *other* firms in the sample in each year, and use it as an instrument in IV regressions. The idea is that the average (market-wide but excluding firm *j*) *II_count* should impact *II_count* in firm *j* but not firm's *j* liquidity. The IV regression results are broadly consistent but the F-statistic of the excluded variable is very sensitive to the model specification used, taking very low values in several cases, which raises concerns about the validity of the instrument in our setting. Hence we do not report these results.

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Table 1
Descriptive Statistics
Panel A. Spread and commonality

	Pre-Crisis Period (2004-6)			Crisis Period (2007-8)			Diff
	N	Mean	Std. Dev	N	Mean	Std. Dev	
Quoted Spread (%)	3,508	0.157	0.197	2,044	0.235	0.522	0.078 [†]
Liquidity Adj R ²	3,498	0.069	0.217	2,038	0.415	0.348	0.346 [†]
Liquidity Beta	3,498	0.955	3.764	2,038	1.282	1.563	0.327 [†]

Note: Quoted Spread is the annualized proportional spread, based on daily closing bid-ask prices. We follow the specification in Chordia et al. (2000) and run the following time-series regressions, using monthly observations for each firm in our sample to calculate the Liquidity Adj. R² and Liquidity Beta:

$$\Delta \text{Spread}_{f,m} = \alpha + \beta_1 \Delta \text{Spread}_{M,m} + \beta_2 \Delta \text{Spread}_{M,m+1} + \beta_3 \Delta \text{Spread}_{M,m-1} + \delta_1 \text{Return}_{M,m} + \delta_2 \text{Return}_{M,m+1} + \delta_3 \text{Return}_{M,m-1} + \delta_4 \Delta \text{Volatility}_{f,m} + \varepsilon_{f,m}$$

where Δ denotes the proportional change in the variable between successive months. The information on the market (M) is calculated using the equal-weighted average (of Spread or Return) for all firms in our sample except firm f . Firm return volatility is calculated for the month m using daily information. We run these time-series regressions separately for years 2004-2006 (pre-crisis) and 2007-2008 (crisis). We define Liquidity Beta as the contemporaneous coefficient of the changes in the market portfolio spread (i.e., β_I). [†] denotes significance of the differences in Means at the 1% level.

Panel B.

Pre- and Crisis period average values of institutional holdings and number of institutional investors

	<i>Institutional Holdings</i>			<i>II_count</i>		
	Pre-Crisis	Crisis	Wilcoxon Rank-sum	Pre-Crisis	Crisis	Wilcoxon Rank-sum
All	0.745	0.787	†	0.232	0.261	†
BNK	0.137	0.145	†	0.004	0.005	†
INS	0.036	0.039	†	0.002	0.002	†
IA	0.440	0.503	†	0.125	0.158	†
PPS	0.028	0.024	†	0.000	0.001	†
UFE	0.001	0.001	†	0.001	0.000	†
QIX	0.475	0.490	†	0.157	0.163	
TRA	0.164	0.178	†	0.066	0.075	†
DED	0.105	0.095	†	0.007	0.006	†

Note: *Institutional holdings* is the ratio of the number of shares held by institutional investors to the total shares outstanding. *II_count* is the number (in thousands) of institutional investors holding a firm's shares. We follow Bushee (2001) and classify institutional investors into the following types and investment styles: bank trusts (BNK), insurance companies (INS), independent advisors (IA), public pension funds (PPS), university and endowment foundations (UFE), quasi-indexers (QIX), transient (TRA), and dedicated (DED). [†] denotes the Wilcoxon rank-sum test significance at the 1% level.

Panel C. Pre- and Crisis period average values of firm characteristics

	Pre-Crisis	Crisis	Wilcoxon Rank-sum
Stock Returns (%)	15.233	-18.313	†
Return Volatility (%)	31.019	53.401	†
Stock Dollar Volume	1.284	2.412	†
Stock Price (\$)	37.530	31.285	†
Assets (\$ mil)	18,252	23,648	†
Leverage	0.192	0.212	†
No of Shareholders (000s)	30.360	32.431	
E-index	2.295	3.615	†
Analyst Coverage	9.969	9.553	

Note: 'Stock Returns' is the annualized average of daily returns. 'Return Volatility' is computed as the standard deviation of daily returns over a year. 'Stock Dollar Volume' is the average daily dollar volume over the year. 'Stock Price' is the average of daily closing prices over the year. 'Assets' is the book value of the firm's assets. Leverage is the ratio of long-term debt to total assets. 'No of Shareholders' is the number of shareholders reported in Compustat. 'E-index' is the Bebchuk et al. (2009) entrenchment index, based on six shareholder rights provisions. Higher values of the E-index are associated with higher levels of managerial entrenchment. 'Analyst Coverage' is the number of one-year-ahead earnings forecasts issued by analysts per year. The information is collected through Datastream. † denotes the Wilcoxon rank-sum test significance at the 1% level.

Table 2
Summary Statistics of Institutional Herding for the Pre-Crisis (2004-6) and Crisis (2007-8) Periods
Panel A. Median and standard deviations of institutional herding

	Median		K-sample test	Standard Deviations		
	Pre-Crisis	Crisis		Pre-Crisis	Crisis	F test
All	0.564	0.486	†	0.215	0.085	†
BNK	0.489	0.460	†	0.211	0.108	†
INS	0.537	0.458	†	0.237	0.122	†
IA	0.572	0.506	†	0.216	0.082	†
PPS	0.500	0.425	†	0.244	0.146	†
UFE	0.583	0.292	†	0.305	0.321	**
QIX	0.536	0.460	†	0.214	0.092	†
TRA	0.599	0.533	†	0.228	0.090	†
DED	0.458	0.449		0.235	0.168	†

Note: Institutional herding is defined as the ratio of buyers to active institutions (buyers plus sellers). A value above (below) 0.5 indicates buy- (sell-) side herding. K-sample and F tests are used to identify the statistical significance of the differences between the two periods. †, ** denote significance at the 1% and 5% levels, respectively. We follow Bushee (2001) and classify institutional investors into the following types and investment styles: bank trusts (BNK), insurance companies (INS), independent advisors (IA), public pension funds (PPS), university and endowment foundations (UFE), quasi-indexers (QIX), transient (TRA), and dedicated (DED).

Panel B.

Pre-crisis period pair-wise correlations of institutional herding between institutional investor groups

	Full sample	BNK	INS	IA	PPS	UFE	QIX	TRA	Average
Full sample									0.89
BNK	0.96								0.85
INS	0.94	0.90							0.85
IA	0.99	0.93	0.92						0.88
PPS	0.90	0.86	0.87	0.87					0.82
UFE	0.54	0.52	0.55	0.52	0.56				0.52
QIX	0.99	0.96	0.94	0.98	0.90	0.55			0.88
TRA	0.98	0.92	0.91	0.98	0.86	0.50	0.95		0.86
DED	0.82	0.78	0.77	0.82	0.73	0.43	0.80	0.80	0.74

Panel C.

Crisis period pair-wise correlations of institutional herding between institutional investor groups

	Full sample	BNK	INS	IA	PPS	UFE	QIX	TRA	Average
Full sample									0.71
BNK	0.86								0.61
INS	0.71	0.63							0.55
IA	0.96	0.74	0.62						0.65
PPS	0.70	0.60	0.58	0.61					0.53
UFE	0.16	0.17	0.16	0.12	0.16				0.13
QIX	0.95	0.88	0.74	0.88	0.72	0.19			0.68
TRA	0.85	0.66	0.60	0.87	0.58	0.07	0.70		0.58
DED	0.44	0.37	0.33	0.44	0.32	0.05	0.41	0.34	0.34

Table 3
Spread and Institutional Investors
Panel A. Full sample regression results for liquidity spread (%)

	Pre-Crisis (2004-6)		Crisis (2007-8)	
II_count	0.216 [†] [0.00]	0.210 [†] [0.00]	0.109 [†] [0.00]	0.095 [†] [0.00]
Institutional holdings	-0.097 [†] [0.00]	-0.098 [†] [0.00]	-0.064 [†] [0.000]	-0.064 [†] [0.00]
Stock returns	0.001 [†] [0.00]	0.001 [†] [0.00]	0.001 [†] [0.00]	0.000 [†] [0.00]
Return volatility (cdf)	0.085 [†] [0.00]	0.097 [†] [0.00]	0.073 [†] [0.00]	0.077 [†] [0.00]
Stock dollar volume (ln)	-0.060 [†] [0.00]	-0.063 [†] [0.00]	-0.039 [†] [0.00]	-0.044 [†] [0.00]
Stock price (ln)	-0.100 [†] [0.00]	-0.100 [†] [0.00]	-0.066 [†] [0.00]	-0.068 [†] [0.00]
Assets (ln)		0.006 [0.12]		0.005 [0.15]
Leverage		0.007 [0.68]		0.017 [0.25]
No of Shareholders (ln)		-0.001 [0.81]		0.001 [0.31]
E-Index		0.001 [0.46]		-0.002 [0.13]
Analyst Coverage		0 [0.56]		0.001 [0.34]
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Number of firm-years	2,608	2,608	1,590	1,590
R ²	0.55	0.54	0.55	0.53

Panel B.

Regression results for liquidity spread (%) for each institutional investor group

		Pre-crisis (2004-6)			Crisis (2007-8)		
		II_count	Institutional holdings	R^2	II_count	Institutional holdings	R^2
i)	BNK	12.362 [†] [0.00]	-0.262 [†] [0.00]	0.56	3.922 [†] [0.00]	-0.206 [†] [0.00]	0.52
ii)	INS	17.626 [†] [0.00]	-0.271 [†] [0.00]	0.55	13.587 [†] [0.00]	-0.251 [†] [0.00]	0.54
iii)	IA	0.292 [†] [0.00]	-0.067 [†] [0.00]	0.56	0.113 [†] [0.00]	-0.066 [†] [0.00]	0.53
iv)	PPS	-1,734 [0.38]	-0.441 [0.52]	0.53	14.897 [†] [0.00]	-0.754 [†] [0.00]	0.52
v)	UFE	11.488* [0.06]	-0.674 [†] [0.00]	0.53	35.507 [†] [0.00]	-0.309 [0.34]	0.52
vi)	QIX	0.252 [†] [0.00]	-0.123 [†] [0.00]	0.53	0.134 [†] [0.00]	-0.072 [†] [0.01]	0.53
vii)	TRA	1.109 [†] [0.00]	-0.187 [†] [0.00]	0.53	0.640 [†] [0.00]	-0.126 [†] [0.00]	0.53
viii)	DED	5.783 [†] [0.00]	-0.083** [0.01]	0.51	3.189 [†] [0.00]	-0.011 [0.63]	0.52

Note: This table reports results from pooled OLS regressions run separately for the pre-crisis (2004-6) and crisis (2007-8) periods. The dependent variable is the firm's quoted spread. All variables are defined as in Table 1. Industry and year fixed effects are included in all specifications. Standard errors are clustered at the firm level. p-values are reported in brackets. †, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Panel A presents the regression results for the full sample. Panel B reports only (to conserve space) the coefficients for 'II_count' and 'Institutional holdings', as well as the regression R^2 , from regressions run separately for each institutional investor type and style.

Table 4
Regression Models on Change of Liquidity Spread, Δ Spread (%)

	Pre-Crisis (2004-6)		Crisis (2007-8)	
Δ II_count	0.168 [†] [0.01]	0.161** [0.01]	0.120 [†] [0.00]	0.089 [†] [0.00]
Δ Institutional holdings	-0.055** [0.02]	-0.049** [0.04]	-0.052** [0.02]	-0.029** [0.04]
Δ Spread (%) _(t-1)		0.071** [0.01]		0.591 [†] [0.00]
Δ Stock returns	0.000 [†] [0.00]	0.000 [†] [0.00]	0.000 [†] [0.00]	0.000 [†] [0.00]
Δ Return volatility (cdf)	0.108 [†] [0.00]	0.110 [†] [0.00]	0.013 [0.18]	0.014 [†] [0.01]
Δ Stock dollar volume (ln)	-0.157 [†] [0.00]	-0.160 [†] [0.00]	-0.053 [†] [0.00]	-0.019** [0.01]
Δ Stock price (ln)	-0.113 [†] [0.00]	-0.117 [†] [0.00]	-0.077 [†] [0.00]	-0.036 [†] [0.00]
Δ Assets (ln)	0.030** [0.04]	0.029** [0.04]	-0.019** [0.01]	-0.009* [0.07]
Δ Leverage	0.007 [0.84]	0.007 [0.83]	-0.024 [0.28]	-0.018 [0.22]
Δ No of Shareholders (ln)	0 [1.00]	-0.001 [0.88]	0.001 [0.70]	-0.004* [0.09]
Δ E-Index	-0.009 [0.40]	-0.007 [0.51]	-0.001 [0.42]	0 [0.73]
Δ Analyst Coverage	0.002** [0.03]	0.002* [0.06]	0 [0.66]	-0.001** [0.02]
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Number of firm-years	2,308	2,284	1,527	1,512
R ²	0.51	0.52	0.25	0.73

Note: This table reports results from pooled OLS regressions on the annual changes in a firm's quoted spread against annual changes of all independent variables. We run separate regressions for the pre-crisis (2004-6) and crisis (2007-8) periods. All variables are defined as in Table 1. In the second and fourth models, we also include the lagged dependent variable. Industry and year fixed effects are included in all specifications. Standard errors are clustered at the firm level. p-values are reported in brackets. †, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

Table 5
Institutional Investors and Commonality in Liquidity during the Crisis Period (2007-8)

Panel A. Full sample test

	Liquidity R^2	Liquidity Beta
II_count	0.230 [†] [0.00]	1.011 [†] [0.01]
Institutional holdings	-0.077 [0.18]	-0.058 [0.82]
Assets (ln)	0.004 [0.71]	0.095** [0.05]
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Number of firm-years	2,038	2,045
R^2	0.08	0.17

Panel B. Regression results for each institutional investor group

		<i>Liquidity R^2</i>			<i>Liquidity Beta</i>		
		II_count	Institutional holdings	R^2	II_count	Institutional holdings	R^2
i)	BNK	9.561** [0.02]	-0.132 [0.48]	0.06	50.44** [0.01]	-0.042 [0.95]	0.16
ii)	INS	21.46 [†] [0.00]	-0.367 [0.26]	0.07	116.0 [†] [0.00]	0.557 [0.71]	0.17
iii)	IA	0.257 [†] [0.01]	-0.092 [0.11]	0.07	1.207** [0.01]	-0.165 [0.46]	0.16
iv)	PPS	8.502 [0.47]	-0.974 [0.27]	0.06	173.2 [†] [0.00]	-3.011 [0.28]	0.17
v)	UFE	46.90** [0.03]	-1.557 [0.49]	0.06	316.6 [†] [0.00]	-3.042 [0.82]	0.16
vi)	QIX	0.341 [†] [0.00]	-0.014 [0.86]	0.08	1.460 [†] [0.00]	0.603* [0.10]	0.17
vii)	TRA	0.905 [†] [0.01]	-0.312 [†] [0.01]	0.08	4.125** [0.03]	-1.389 [†] [0.00]	0.17
viii)	DED	5.389 [0.15]	-0.144 [0.27]	0.07	31.995* [0.06]	-1.114** [0.04]	0.17

Note: This table reports results from pooled OLS regressions on the influence of institutional investors on liquidity risk during the crisis (2007-8) period. The dependent variables are the two proxies for liquidity risk used in this study, i.e., Liquidity Adj R^2 and Liquidity Beta. In Panel A (Panel B) we run regressions for the full sample (separately by each institutional investor type/style). Panel B reports only the coefficients for 'II_count' and 'Institutional holdings', as well as the regression R^2 , to conserve space. All variables are defined as in Table 1. Industry and year fixed effects are included in all specifications. Standard errors are clustered at the firm level. p-values are reported in brackets. [†], ** and * denote significance at the 1%, 5% and 10% levels, respectively.

Table 6
Institutional Herding and Market Liquidity during the Crisis Period (2007-8)

		Spread (%)		Liquidity R ²		Liquidity Beta	
		<i>coef.</i>	<i>R</i> ²	<i>coef.</i>	<i>R</i> ²	<i>coef.</i>	<i>R</i> ²
i)	Full sample	-0.279 [†] [0.00]	0.52	-0.403** [0.03]	0.06	-2.594 [†] [0.00]	0.17
ii)	BNK	-0.134 [†] [0.00]	0.52	-0.207* [0.06]	0.06	-0.780* [0.10]	0.16
iii)	INS	0.044 [0.21]	0.51	-0.142* [0.10]	0.06	-0.572* [0.07]	0.16
iv)	IA	-0.306 [†] [0.00]	0.52	-0.296* [0.08]	0.06	-2.460 [†] [0.00]	0.17
v)	PPS	-0.100 [†] [0.00]	0.52	-0.174** [0.01]	0.06	-0.597** [0.04]	0.16
vi)	UFE	-0.005 [0.35]	0.55	0.024 [0.42]	0.07	-0.022 [0.85]	0.15
vii)	QIX	-0.162 [†] [0.00]	0.51	-0.248* [0.10]	0.06	-1.629 [†] [0.01]	0.16
viii)	TRA	-0.230 [†] [0.01]	0.52	-0.377** [0.01]	0.06	-1.383 [†] [0.01]	0.16
ix)	DED	-0.011 [0.49]	0.50	0.038 [0.47]	0.05	-0.083 [0.66]	0.16

Note: This table reports results from pooled OLS regressions on the impact of institutional herding on quoted spreads and liquidity commonality during the crisis (2007-8) period. We use the model specifications presented in Table 3 (Table 5) when the dependent variable is Spread (Liquidity R² or Liquidity Beta), but we replace the 'II_count' and 'Institutional holdings' with 'Institutional herding'. We only report the coefficients for herding and the R² of each regression to conserve space. All variables are defined as in Table 1. Industry and year fixed effects are included in all specifications. Standard errors are clustered at the firm level. p-values are reported in brackets. †, ** and * denote significance at the 1% and 5% and 10% levels, respectively.

Table 7
Sell-Side Institutional Herding, Degree of Imbalance and Market Liquidity during the Crisis Period (2007-8)

		Spread (%)		Liquidity R ²		Liquidity Beta	
		<i>coef.</i>	<i>R</i> ²	<i>coef.</i>	<i>R</i> ²	<i>coef.</i>	<i>R</i> ²
i)	Full sample	0.671 [†] [0.00]	0.52	0.992** [0.02]	0.06	5.693 [†] [0.00]	0.16
ii)	BNK	0.348 [†] [0.00]	0.52	0.447 [0.12]	0.06	1.978* [0.10]	0.16
iii)	INS	-0.233** [0.03]	0.52	0.339 [0.15]	0.06	0.821 [0.36]	0.16
iv)	IA	0.686 [†] [0.00]	0.52	0.711* [0.10]	0.06	5.725 [†] [0.00]	0.17
v)	PPS	0.183 [†] [0.01]	0.52	0.307* [0.06]	0.06	0.425 [0.55]	0.16
vi)	UFE	0.012 [0.58]	0.55	-0.050 [0.71]	0.07	0.363 [0.54]	0.15
vii)	QIX	0.400 [†] [0.00]	0.52	0.514 [0.18]	0.06	3.486 [†] [0.01]	0.16
viii)	TRA	0.387 [†] [0.01]	0.52	0.530 [0.11]	0.06	2.546* [0.08]	0.16
ix)	DED	-0.022 [0.64]	0.50	-0.270* [0.08]	0.05	-0.786 [0.16]	0.16

Note: This table reports results from pooled OLS regressions on the impact of sell-side institutional herding on quoted spreads and liquidity commonality during the crisis (2007-8) period. We use the model specifications presented in Table 3 (Table 5) when the dependent variable is Spread (Liquidity R² or Liquidity Beta), but we replace the 'II_count' and 'Institutional holdings' with a new variable, which is the interaction of a sell-side herding dummy (that takes the value of one when institutional herding is lower or equal to 0.5, zero otherwise) and a Degree of Imbalance (DOI) variable (that is defined as the absolute value of the difference of institutional herding from 0.5). We also include in the models the uninteracted terms. We only report the coefficients for the interaction term and the R² of each regression to conserve space. All other variables are defined as in Table 1. Industry and year fixed effects are included in all specifications. Standard errors are clustered at the firm level. p-values are reported in brackets. †, ** and * denote significance at the 1% and 5% and 10% levels, respectively.